TECHNICAL DILIGENCE REVIEW - RECORD 020

Helioscope Robotics, Inc. - diligence file reference TECHNICAL_DILIGENCE-020

1. Scope of the safety review and the documents examined

The subject of this report is the functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet, examined for Ardent Peak by Brandywine Technology Partners over 3 weeks ending 2026-05-21. This report was prepared by Brandywine Technology Partners for Ardent Peak and covers the functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet at Helioscope Robotics, Inc. as at 2026-05-21. N. Thackeray was granted read access to the code, the pipeline and the incident tracker for the purposes of this report; production access was not requested. Access for the present report was sufficient for the conclusions drawn; where it was not, the limitation is stated at the point it bites. Brandywine Technology Partners was given read access without restriction on scope within the agreed subject matter.

Evidence gathering for this report was iterative: an initial document request, 17 interviews, then a second request that closed most of the gaps. The working method behind the report at TDR-2026-22 was to start from the incident record and work back to the design, rather than the other way round. Two document requests remained partly unsatisfied when TDR-2026-22 was delivered, and the affected conclusions are marked. One limitation of report TDR-2026-22 should be stated plainly: the reviewer had no production access and relied on the company's own measurements for live latency. N. Thackeray interviewed 17 people for report TDR-2026-22; where an account was uncorroborated it is marked as such.

9 teams contributed to the interview programme for report TDR-2026-22, and one of them was interviewed twice after a document arrived late. 17 interviews inform the report at TDR-2026-22. Where an interview is the sole source for a claim, the claim is graded limited. The instruction behind the present report was narrow: examine the functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet and report by 2026-05-21. This report, this report, is the Brandywine Technology Partners workstream on the functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet, delivered to Ardent Peak on 2026-05-21. TDR-2026-22 deliberately preserves detail that a summary would drop, because the detail is where the qualifications live.

This report carries one structural limitation: the company's own metrics are the only source for several objectives, and the reviewer could not build an independent measurement inside the engagement. The principal limitation on this report is time: 3 weeks was enough to test the control path and not enough to test the reporting path. The evidence convention in TDR-2026-22 is that severity describes consequence and evidence strength describes confidence, and the two are never merged. Nothing in TDR-2026-22 is presented as verified on the strength of an interview alone. The functional-safety analysis, hazard register and protective-stop behaviour of the robot fleet produced more evidence than the present report reproduces, and the working papers hold the remainder.

The present report distinguishes three positions: verified defect, credible concern, and matter requiring further work. The third category is not empty. N. Thackeray has left several concerns in the present report at limited evidence strength rather than dropping them or dressing them up. Nothing in this report rests on a test the reviewer designed; the tests were the company's, and the reviewer read the results and the method. The report at TDR-2026-22 is limited by log retention: the telemetry window is ninety days, so events before 2025-11-15 could not be examined directly. N. Thackeray has kept the working papers behind TDR-2026-22 in a form that a second reviewer could re-perform.

Report TDR-2026-22 sampled 19 services from an inventory of 34, weighted towards those on the critical control path. Sampling in the report at TDR-2026-22 covered 19 of the 34 catalogued services and 6 of the 18 live sites. Minutes of every the report at TDR-2026-22 interview were returned to the participant, and three came back with substantive corrections. The interview programme behind the present report covered 9 teams and produced 10 follow-up requests, of which most were satisfied. Brandywine Technology Partners raised each finding with the owning engineer before it was written up, and the responses are recorded.

Schedule I - Interview programme and follow-up requests arising (TDR-2026-22)

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
2025-08-05	Nathan Oduya	security engineer	Motion Planner	2	1
2025-10-12	Nadia Semenova	security engineer	Telemetry Ingest	3	2
2026-03-06	Yuki Harada	quality lead	Order Broker	3	0

Date	Participant	Role	Principal topic	Follow-ups raised	Follow-ups open
2025-11-24	Priya Raghunathan	principal engineer	Battery Management Service	5	1
2026-02-12	Yuki Harada	head of engineering	Motion Planner	2	2
2026-08-20	Nadia Semenova	site reliability lead	Fleet Allocator	4	2
2026-05-26	Anjali Sekhon	team lead	Battery Management Service	3	2
2025-08-30	Bettina Vogel	quality lead	Charge Scheduler	1	0
2026-09-25	Nathan Oduya	principal engineer	Payload Scale Interface	2	1
2025-10-31	Grace Lindqvist	security engineer	Safety Controller	2	2
2026-05-11	Grace Lindqvist	field reliability lead	Gripper Controller	4	1
2025-12-19	Marcus Duplessis	site reliability lead	Event Bus	1	0
2026-06-07	Tomas Villareal	safety engineer	Site Provisioning Service	3	3
2026-06-02	Nathan Oduya	quality lead	Battery Management Service	5	1
2025-08-21	Anjali Sekhon	release manager	Fleet Health Monitor	3	2
2026-02-12	Wanda Njoku	principal engineer	Analytics Warehouse Loader	1	0
2026-01-19	Farida Cisse	head of engineering	Release Orchestrator	5	5
2025-09-02	Nadia Semenova	principal engineer	Safety Controller	1	1
2025-12-08	Lucia Ferrante	engineering owner	Billing Usage Meter	1	0
2025-09-26	Rahul Menon	engineering owner	Motion Planner	1	1
2026-05-31	Ines Duarte	site reliability lead	Traffic Manager	3	1
2026-08-03	Tomas Villareal	security engineer	Event Bus	5	5
2026-09-05	Sam Oyelaran	quality lead	Order Broker	1	0
2026-04-02	Farida Cisse	principal engineer	Telemetry Ingest	4	1
2025-11-11	Farida Cisse	field reliability lead	Event Bus	4	2
2025-12-27	Marta Kellerman	release manager	Robot Shadow Store	4	1
2025-12-06	Theo Bassett	engineering owner	OTA Updater	4	1
2025-10-19	Sam Oyelaran	safety engineer	Billing Usage Meter	1	0
2026-03-24	Bettina Vogel	head of engineering	Map Authoring Studio	2	1
2025-11-11	Priya Raghunathan	security engineer	OTA Updater	4	2

2. Hazard identification: HZ-13 and the register as a whole

The register places at HZ-13 the scenario in which robot ignores a temporary barrier placed by site staff. Serious injury harm, Occasional exposure and Medium controllability are the parameters recorded for register entry HZ-13. Occasional-exposure hazard HZ-13 was verified through obstacle detection bench test plus a site procedure review. hazard HZ-13 in the register illustrates why TDR-2026-22 separates the severity of a finding from the confidence behind it.

Where hazard HZ-17 in the register bites is the newest three sites, where commissioning is incomplete. Hazard HZ-17 in the register is not clean: extraction runs weekly, so a busy robot can still lose events between extractions. On history, hazard HZ-17 has produced no field event in the reviewed period. No direct harm hazard HZ-17 is one of the entries where TDR-2026-22 relies on the company's own measurement rather than an independent one.

Minor injury hazard HZ-14 produced two near-miss reports from site staff; that is worth stating alongside the assessed exposure. Hazard HZ-14 in the register is mitigated by explicit robot selection with a live video confirmation before motion. HZ-14 was tested by console interaction test plus a review of session records, and the reviewer read the resulting record. hazard HZ-14 would be a candidate for the first post-close engineering review under any reasonable plan.

Every robot variant currently in production is the exposed population for Remote-exposure hazard HZ-21. Verification of HZ-21 was by witnessed push test on two robot variants. Broadly acceptable is the residual position on hazard HZ-21, whose status is verified. hazard HZ-21 in the register was one of two items where TDR-2026-22 could not obtain the underlying data within the engagement.

The verification method for HZ-19 is occupancy sensor bench test plus a witnessed dock approach. On hazard HZ-19 in the register the reviewer records that occupancy sensing is fitted at every dock revision currently deployed. HZ-19 covers the case where robot enters a dock occupied by a person. HZ-19 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Mitigation for hazard HZ-02 in the register is aisle-end speed reduction plus a mirrored sensing field at mapped blind corners. The population exposed to Serious injury hazard HZ-02 is the sites that have moved to opportunity charging. The parameter set for HZ-02 reads Serious injury, Occasional, Low. hazard HZ-02 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

Hazard HZ-15 in the register carries residual risk of Tolerable with a status of partially verified. Serious injury hazard HZ-15 has produced no event, though the control has activated repeatedly; that is worth stating alongside the assessed exposure. Where HZ-15 bites is every site with a mezzanine or ramp. hazard HZ-15 was reviewed against the position twelve months earlier, and nothing measurable has changed.

Equipment damage hazard HZ-16 was verified through two rollback drills plus a fleet version audit. Produced two near-miss reports from site staff is the incident history behind HZ-16. Hazard HZ-16 in the register covers the case where firmware update leaves a robot in a mixed-version state. hazard HZ-16 in the register was checked against the corresponding entry in the company's own register, and the register was current.

Schedule N - Hazard identification and risk assessment register (TDR-2026-22)

Ref	Hazard	Severity	Exposure	Controllability	Mitigation	Verification method	Residual risk
HZ-01	Robot contacts a pedestrian while travelling in a shared aisle	Serious injury	Frequent	Low	speed and separation monitoring with a protective stop on margin breach	witnessed approach testing on six robots plus a review of the on-robot safety log	Tolerable with the measured approach speed assumption stated
HZ-02	Robot contacts a pedestrian emerging from a blind cross-aisle	Serious injury	Occasional	Low	aisle-end speed reduction plus a mirrored sensing field at mapped blind corners	site-specific commissioning test at each mapped blind corner	Tolerable at commissioned sites; two sites have unmapped corners
HZ-03	Load falls from an elevated deck during transport	Serious injury	Occasional	Medium	payload mass and centre-of-gravity limits enforced by the scale interface	bench drop testing plus a payload limit audit across the fitted fleet	Tolerable where the weighing deck is fitted
HZ-04	Gripper closes on a hand during a manual intervention	Serious injury	Remote	Low	force limiting with a mechanical release and an interlock on the access panel	force bench test after every calibration, evidenced by a calibration record	Tolerable once calibration is gated in the service workflow
HZ-05	Drive power is not removed when the emergency stop is actuated	Fatality	Improbable	Low	hardwired dual-channel stop chain independent of the compute module	witnessed actuation test on every unit at build and at annual service	Broadly acceptable
HZ-06	Robot moves unexpectedly during maintenance	Serious injury	Occasional	Low	maintenance mode with a physical lockout and a two-step exit confirmation	procedure walk-through with three service technicians and a lockout inventory	Tolerable
HZ-07	Robot travels into an unmapped area and contacts fixed plant	Equipment damage	Occasional	Medium	map keep-out enforcement plus a boundary confidence check in localization	simulation coverage of boundary cases plus a site survey audit	Tolerable where survey quality is confirmed
HZ-08	Two robots collide at a corridor intersection	Equipment damage	Frequent	High	corridor reservation with deadlock detection and a deterministic yield tie-break	reservation protocol replay plus the pinch-point simulation scenario	Tolerable

Ref	Hazard	Severity	Exposure	Controllability	Mitigation	Verification method	Residual risk
HZ-09	Robot fails to stop on a descending ramp when a wheel loses traction	Serious injury	Remote	Medium	ramp speed limits, a slip detector and a mechanical brake hold	ramp testing on a wet and a dry surface at two sites	Not yet tolerable without the slip detector
HZ-10	Charging contact arcs during an unplanned disconnect	Fire	Improbable	Low	contactor sequencing with an arc-suppression design in the charge dock	supplier test report plus an internal witnessed disconnect test	Broadly acceptable
HZ-11	Battery pack thermal runaway during opportunity charging	Fire	Improbable	Low	cell temperature limits, pack-level current derating and a dock thermal cutout	supplier abuse test report plus a review of pack fault records	Broadly acceptable, subject to the state-of-health drift item
HZ-12	Operator is struck by a robot reversing out of a charge dock	Minor injury	Occasional	Low	reverse motion audible warning plus a rear sensing field	witnessed reverse test plus a review of the dock layout at four sites	Tolerable
HZ-13	Robot ignores a temporary barrier placed by site staff	Serious injury	Occasional	Medium	detection of unmapped obstacles with a stop, plus a site procedure for barriers	obstacle detection bench test plus a site procedure review	Tolerable where the barrier meets the visibility guidance
HZ-14	Remote operator issues a motion command to the wrong robot	Minor injury	Occasional	Low	explicit robot selection with a live video confirmation before motion	console interaction test plus a review of session records	Tolerable once the stale roster defect is closed
HZ-15	Robot obstructs an emergency egress route	Serious injury	Remote	Medium	egress routes marked as keep-out and a stop-and-clear behaviour on a fire signal	site commissioning check plus a fire drill observation at two sites	Tolerable
HZ-16	Firmware update leaves a robot in a mixed-version state	Equipment damage	Remote	Low	atomic update with rollback and a version consistency check before mission acceptance	two rollback drills plus a fleet version audit	Tolerable
HZ-17	Safety controller fault log overwrites evidence needed for an investigation	No direct harm	Frequent	Low	periodic fault log extraction to the cloud with a retention policy	log extraction record review across the fleet	Tolerable, with an evidence-loss caveat
HZ-18	Speed reduction engages too late for a fast-approaching person	Serious injury	Occasional	Low	protective distance calculation with a stated approach speed assumption	bench approach testing at the assumed speed and at a higher speed	Not yet tolerable at the higher approach speed
HZ-19	Robot enters a dock occupied by a person	Serious injury	Remote	Low	dock occupancy sensing plus a low approach speed inside the dock envelope	occupancy sensor bench test plus a witnessed dock approach	Tolerable
HZ-20	Lidar	Serious injury	Occasional	Medium	return-intensity	contamination	Tolerable

Ref	Hazard	Severity	Exposure	Controllability	Mitigation	Verification method	Residual risk
	contamination degrades detection without a fault being raised				monitoring with a cleaning prompt and a degraded-mode speed limit	bench test plus a review of cleaning compliance at three sites	where cleaning compliance is evidenced
HZ-21	Manual push of a powered robot causes an uncontrolled movement	Minor injury	Remote	Low	push detection that engages the brake and requires a deliberate release	witnessed push test on two robot variants	Broadly acceptable
HZ-22	Site network loss leaves robots operating without fleet coordination	Equipment damage	Occasional	Medium	local yield behaviour with a conservative speed limit and a timed stop	network loss injection at one site plus a bench network partition test	Tolerable for outages under the tested duration

3. Risk assessment parameters and residual risk

Remote-exposure hazard HZ-09 is not clean: the slip detector is prototyped and unvalidated as at the review date. Against register entry HZ-09 the team relies on ramp speed limits, a slip detector and a mechanical brake hold. The three risk terms on Remote-exposure hazard HZ-09 are severity Serious injury, exposure Remote and controllability Medium. hazard HZ-09 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

HZ-12 was verified through witnessed reverse test plus a review of the dock layout at four sites. Register entry HZ-12 applies to every robot variant currently in production. Residual risk at hazard HZ-12 in the register is stated as Tolerable, and the status is verified. hazard HZ-12 in the register was tested rather than accepted, and the test is described in the working papers behind TDR-2026-22.

Register entry HZ-22 applies to the newest three sites, where commissioning is incomplete. Reading HZ-22 against the verification evidence, the tested duration is shorter than the five-hour outage seen in April 2026. Has never activated in the field and is evidenced only on the bench is the incident history behind Equipment damage hazard HZ-22. Equipment damage hazard HZ-22 rests on a single source, and the reviewer has said so rather than implying corroboration.

What the register does not say about Improbable-exposure hazard HZ-05, and the reviewer does, is that nine field units retain a pre-production harness revision and are tracked to retrofit. Field history for hazard HZ-05 is that it has produced no event, though the control has activated repeatedly. HZ-05 applies to the manipulator-equipped variant only. hazard HZ-05 was one of two items where TDR-2026-22 could not obtain the underlying data within the engagement.

Occasional-exposure hazard HZ-18 carries an open qualification: a design study to measure closing rate is running with a vendor assessment booked. Occasional-exposure hazard HZ-18 was verified through bench approach testing at the assumed speed and at a higher speed. The parameter set for HZ-18 reads Serious injury, Occasional, Low. hazard HZ-18 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-22.

Produced a single event before the current control was fitted is the incident history behind hazard HZ-06 in the register. Reading Serious injury hazard HZ-06 against the verification evidence, one site had two lockout devices for eleven robots at the time of the visit. Hazard HZ-06 is mitigated by maintenance mode with a physical lockout and a two-step exit confirmation. HZ-06 rests on a single source, and the reviewer has said so rather than implying corroboration.

The hazard recorded at Occasional-exposure hazard HZ-07 is that robot travels into an unmapped area and contacts fixed plant. Residual risk at hazard HZ-07 in the register is stated as Tolerable where survey quality is confirmed, and the status is partially verified. For Occasional-exposure hazard HZ-07 the verifying activity was simulation coverage of boundary cases plus a site survey audit. hazard HZ-07 is one of the entries where TDR-2026-22 relies on the company's own measurement rather than an independent one.

The design measure recorded at HZ-08 is corridor reservation with deadlock detection and a deterministic yield tie-break. Two robots collide at a corridor intersection is the hazard entered at hazard HZ-08 in the register. Equipment

damage harm, Frequent exposure and High controllability are the parameters recorded for register entry HZ-08. HZ-08 was checked against the corresponding entry in the company's own register, and the register was current.

4. Safety-related subsystems and their verification

10 defects remain open against Battery Management Service; the one the reviewer pressed on is that state-of-health estimates drift on packs that spend most of their life on opportunity charge. For Battery Management Service the supportability position is one pack revision uses a cell supplier that has since discontinued the part. Coverage for Battery Management Service was 54 per cent line coverage at the review date, and the trend over the reviewed period was flat, with a one-off drop when a test suite was quarantined. On paper Battery Management Service releases pack firmware twice a year, the cloud service monthly, and the observed practice did not diverge from that. Peter Halvorsen leads Robot Firmware's work on Battery Management Service and was the reviewer's principal interview for it. Battery Management Service is the clearest instance in TDR-2026-22 of a control that exists on paper and is weaker in practice.

Battery Management Service was assessed against the objective and the defect record, and the objective is met in production, but the evidence that it will be met after the next dependency upgrade is weak. Battery Management Service owns its own data and exposes it only through a versioned interface, which the reviewer read. The objective recorded against Battery Management Service is that pack cell imbalance is reported before it reaches the derating threshold, and it is reported weekly to the engineering leadership. Reading the dependency map from Battery Management Service outward gives the pack sensors and the charge scheduler on the input side and the charge scheduler and the fleet health monitor on the output side. Documents and data inspected for Battery Management Service were the pack firmware release notes, cell test data for two revisions and a derating incident file. Battery Management Service was excluded from an earlier draft of TDR-2026-22 and reinstated after the reviewer obtained the supporting record.

Inputs to Config Distribution come from the site provisioning service and the artefact signing service, and its consumers are the on-robot supervisor and the motion planner parameters. Config Distribution sits split edge and cloud, with a signed bundle format, and no alternative placement is currently supported. Config Distribution releases weekly. Test coverage on Config Distribution is 72 per cent line coverage and the measurement basis was confirmed with Sam Oyelaran. Sam Oyelaran leads Platform Infrastructure's work on Config Distribution and was the reviewer's principal interview for it. Config Distribution is reported here in full because a summary of it in an earlier draft of TDR-2026-22 was misread as an assurance.

Config Distribution was assessed on the bundle format specification, distribution latency measurements and an offline apply test, and nothing beyond that was available in the data room. Config Distribution was built in Rust 1.79 agent with a Go 1.22 control service and has not been rewritten; it still runs on split: an on-robot agent and a cloud service against content-addressed object storage with a PostgreSQL 14 index. Data ownership is clear for Config Distribution: it writes content-addressed object storage with a PostgreSQL 14 index and no other service writes there. Config Distribution was tested against the dependency ageing register, which records none; the bundle format was revised in 2025. Platform Infrastructure publishes one objective for Config Distribution: a configuration change reaches every online robot inside five minutes. Config Distribution is the reason one section of TDR-2026-22 concludes narrowly rather than generally.

Video Snippet Service's dependency ageing was checked: a transcoder build with known but low-severity upstream advisories. Interface discipline around Video Snippet Service is adequate; the schema is versioned and change is gated on a compatibility check. The remote operations console and the incident review workflow rely on Video Snippet Service, which in turn relies on the on-robot camera buffer and the identity service. Video Snippet Service is deployed cloud, regional, with an on-robot capture buffer. Video Snippet Service is promoted monthly, with the ring order fixed by policy. Video Snippet Service was one of two items where TDR-2026-22 could not obtain the underlying data within the engagement.

The evidence base for Video Snippet Service was the retention configuration, a lifecycle audit across regions and the access log sample. Video Snippet Service shows 43 per cent line coverage in the pipeline, and that is below what this component's failure mode warrants. Go 1.22 with a media pipeline is the implementation language for Video Snippet Service, which is hosted on cloud, containers plus a media transcoder with object storage with a thirty-day lifecycle and a PostgreSQL 14 index behind it. Read against its objective, Video Snippet Service management's own assessment is

broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. The open defect backlog for Video Snippet Service holds 12 items, the most consequential being that the retention lifecycle was applied per bucket, so one region kept clips for ninety days. Video Snippet Service is the reason one section of TDR-2026-22 concludes narrowly rather than generally.

Against Billing Usage Meter, the coverage figure is 79 per cent line coverage, which the figure is respectable but excludes the fallback path, which is where the known defect sits. Upstream of Billing Usage Meter sit the order broker and the WMS connector; downstream, the finance export and the customer usage report consume its output. Data and Telemetry publishes one objective for Billing Usage Meter: metered usage reconciles to the order journal within 0.5 per cent each month. The backlog for Billing Usage Meter is 7 items deep and contains an acknowledged defect whereby a re-run of a failed metering window can double-count if the operator skips the reset step. Nadia Semenova of Data and Telemetry maintains Billing Usage Meter, and the service inventory entry was current at the review date. Billing Usage Meter is reported at the evidence strength available and has not been promoted to fit the narrative.

Billing Usage Meter sits cloud, regional, and no alternative placement is currently supported. Evidence examined for Billing Usage Meter comprised the metering specification, three monthly reconciliations and the ledger immutability check. Release cadence for Billing Usage Meter is monthly, aligned to the billing cycle, and the release records bear that out. End-of-life review of Billing Usage Meter found none identified. Billing Usage Meter owns its own data and exposes it only through a versioned interface, which the reviewer read. Billing Usage Meter is reported here in full because a summary of it in an earlier draft of TDR-2026-22 was misread as an assurance.

Against Lidar Driver, the coverage figure is 44 per cent line coverage, which the reviewer notes that the untested share includes the error handling. Robot Firmware holds Lidar Driver, and the ownership entry naming Yuki Harada matches the on-call rotation. The sensor hardware and the time synchronisation daemon feed Lidar Driver; a failure in Lidar Driver is felt by the costmap fusion pipeline and the localization engine. Data ownership is clear for Lidar Driver: it writes no store; frames are handed to fusion in memory and no other service writes there. Behind Lidar Driver sits no store; frames are handed to fusion in memory; the service itself is C++17 on on-robot Linux, user-space driver. Lidar Driver has no single accountable owner, and that is the finding the reviewer would press hardest.

The evidence base for Lidar Driver was the driver source, the vendor integration guide and a captured reconnect trace. The open defect backlog for Lidar Driver holds 18 items, the most consequential being that reconnect handling leaves the timestamp offset uncorrected for the first three frames. Service-level intent for Lidar Driver states that no more than one dropped scan per ten thousand at nominal spin rate. The assessment of Lidar Driver is not unqualified: management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. Changes to Lidar Driver reach production as needed, historically twice a year. Lidar Driver is the entry that changed most between the first and second document requests behind TDR-2026-22.

The build system, the simulation harness gate and the identity service feed Release Orchestrator; a failure in Release Orchestrator is felt by the OTA updater and the cloud deployment controllers. Colin Achebe leads Developer Productivity's work on Release Orchestrator and was the reviewer's principal interview for it. On paper Release Orchestrator releases continuous, self-hosting, and the observed practice did not diverge from that. The evidence base for Release Orchestrator was the release policy, ninety release records and an audit of emergency-path usage. Release Orchestrator was examined for unsupported components; none identified. Release Orchestrator was one of two items where TDR-2026-22 could not obtain the underlying data within the engagement.

Release Orchestrator does not share a schema with any consumer; every read by the OTA updater and the cloud deployment controllers goes through the published interface. Release Orchestrator pairs a Go 1.22 implementation with cloud, a pipeline runner with approval hooks, and its data lives in PostgreSQL 14 for release records and object storage for artefacts. 55 per cent line coverage is reported for Release Orchestrator, and the reviewer re-ran the measurement on a clean checkout. On Release Orchestrator the conclusion is qualified: the position is stronger than the coverage figure alone would suggest, because the bench evidence is unusually good here. The manifests place Release Orchestrator cloud, single global deployment. Release Orchestrator was inspected in the repository rather than described from the ticket, which is why the account is specific.

Charge Scheduler was built in Python 3.11 and has not been rewritten; it still runs on cloud, containers on a managed Kubernetes cluster against PostgreSQL 14, shared instance with the allocator. The cadence recorded for Charge

Scheduler is fortnightly, without a formal approval gate until March 2026, which matches the observed release history. Charge Scheduler reports 63 per cent line coverage, measured by the same instrumentation the pipeline uses. Inputs to Charge Scheduler come from the battery management service and the site shift calendar, and its consumers are the fleet allocator and the fleet health monitor. Nadia Semenova leads Fleet Services's work on Charge Scheduler and was the reviewer's principal interview for it. Charge Scheduler sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The assessment of Charge Scheduler is not unqualified: the component is adequate today and carries a supportability liability that is not on the funded plan. Charge Scheduler sits cloud, regional, with a conservative local fallback policy on the robot, and no alternative placement is currently supported. Evidence examined for Charge Scheduler comprised the scheduler policy document and four sites' charge utilisation exports. The interface Charge Scheduler publishes is versioned, and the fleet allocator and the fleet health monitor were checked for use of any undocumented field. The open defect backlog for Charge Scheduler holds 7 items, the most consequential being that opportunity-charge windows are not released when a shift ends earlier than planned. Charge Scheduler was checked against the company's own register, which had not been updated since the change landed.

Schedule A - Service inventory, ownership and objective attainment (TDR-2026-22)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Motion Planner	Rust 1.79	Autonomy Core	71%	plan cycle completes inside 40 ms at the 99th percentile	the costmap fusion pipeline and the localization engine	none; the toolchain is pinned to a supported compiler release
Costmap Fusion	C++17	Perception	58%	fused occupancy grid published at 20 Hz with sensor-to-grid latency under 60 ms	the lidar driver, the depth camera bridge and the wheel odometry filter	a vendored geometry library whose upstream branch left support in 2024
Fleet Allocator	Go 1.22	Fleet Services	82%	task assignment inside 900 ms of order arrival at p95	the order broker and the robot shadow store	none identified in the reviewed manifests
Traffic Manager	Go 1.22	Fleet Services	74%	corridor deadlock detected and broken inside two seconds	the fleet allocator and the site map service	Redis 6 client library still vendored in one code path
Charge Scheduler	Python 3.11	Fleet Services	63%	fleet state of charge stays above 35 per cent through the peak shift at every site	the battery management service and the site shift calendar	none; the dependency set was refreshed in the 2026 platform upgrade
Safety Controller	C, written to a constrained coding subset	Safety Engineering	91%	a validated protective-stop trigger de-energises the drive inside 120 ms	the E-stop chain, the speed and separation monitor and the drive current sensors	none; the supplier's support commitment runs to 2031
E-Stop Chain	VHDL	Safety Engineering	witnessed test	drive power is removed within 250 ms of button actuation on every tested unit	the operator stop buttons and the perimeter interlocks	none; the relay parts are current production
Lidar Driver	C++17	Robot Firmware	44%	no more than one dropped scan per ten thousand at nominal spin rate	the sensor hardware and the time synchronisation daemon	the vendor's software development kit is two major versions behind current
Localization Engine	C++17	Autonomy Core	66%	pose error stays under 25 mm in mapped aisles at survey speed	the lidar driver, the wheel odometry filter and the map distribution channel	an unmaintained scan-matching library is still linked in the fallback path

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
WMS Connector	Java 21	Integrations	69%	order acknowledgement returned to the warehouse system inside 1.5 s at p95	the customer warehouse management systems and the identity service	one customer integration still runs on a connector build from 2024
Order Broker	Go 1.22	Fleet Services	78%	no accepted order is lost, measured by journal reconciliation each night	the WMS connector and the site configuration service	none identified
Telemetry Ingest	Go 1.22	Data and Telemetry	61%	telemetry visible in the operator console inside 30 s of emission at p95	the edge gateway agent and the event bus	the stream processor runs a release that leaves community support in late 2026
OTA Updater	Rust 1.79 on the robot, Go 1.22 in the cloud	Platform Infrastructure	76%	an update is either fully applied or fully rolled back, with no mixed-version robot	the release orchestrator and the artefact signing service	none; artefact signing was moved to the current key format in 2025
Identity and Authorization	Go 1.22	Security Engineering	84%	token issuance inside 200 ms at p95 and no unauthenticated write path	the customer directory federations and the secret store	none identified in the reviewed manifests
Remote Operations Console	TypeScript	Applications	52%	operator input reaches the robot inside 350 ms at p95 on a supported network	the identity service, the telemetry ingest path and the video snippet service	a charting library that no longer receives security updates
Simulation Harness	Python 3.11	Developer Productivity	47%	the nightly scenario suite completes inside four hours	the planner and localization builds under test	the physics core depends on a library branch that stopped receiving fixes in 2023
Hardware-in-the-Loop Rig	Python 3.11 test harness around production firmware	Robot Firmware	39%	every firmware candidate runs the full bench suite before ring one	the firmware candidate artefacts and the bench sensor stimulators	the rig controller runs an operating system release that left vendor support in 2025

4. Safety-related subsystems and their verification - part 2 of 2

A further extract of the 4. Safety-related subsystems and their verification appears here, taken in order from the point at which the previous part ended. The sequence is unbroken either side of it.

Readers should carry the 4. Safety-related subsystems and their verification's stated conventions across the break, since this part adds no definition of its own. Pagination alone determines where it falls.

Schedule A - Service inventory, ownership and objective attainment (TDR-2026-22) (continued, part 2 of 2)

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Fleet Health Monitor	Python 3.11	Field Reliability	57%	a fleet-affecting condition raises an alert inside three minutes of first symptom	the telemetry ingest path and the robot shadow store	none identified
Battery Management Service	C on the pack controller, Python 3.11 in the cloud	Robot Firmware	54%	pack cell imbalance is reported before it reaches the derating threshold	the pack sensors and the charge scheduler	one pack revision uses a cell supplier that has since discontinued the part
Gripper Controller	C++17	Robot Firmware	41%	a grasp completes or reports failure inside 1.2 s	the payload scale interface and the perception grasp proposals	none; the coprocessor part is current production

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
Map Authoring Studio	TypeScript	Applications	49%	a published map revision reaches every robot at the site inside fifteen minutes	the site survey imports and the identity service	a geometry rendering library one major version behind its supported line
Site Provisioning Service	Go 1.22	Platform Infrastructure	64%	a new site reaches a working configuration inside one working day	the identity service and the customer network intake records	none identified
Event Bus	operated open-source broker, no in-house code	Platform Infrastructure	witnessed test	publish acknowledgement inside 25 ms at p95 and no partition without a leader	every publishing service	the broker minor release in production leaves the vendor's support window in 2027
Config Distribution	Rust 1.79 agent	Platform Infrastructure	72%	a configuration change reaches every online robot inside five minutes	the site provisioning service and the artefact signing service	none; the bundle format was revised in 2025
Robot Shadow Store	Go 1.22	Fleet Services	68%	shadow state is no more than two seconds behind the robot at p95	the telemetry ingest path and the edge gateway agent	none identified
Analytics Warehouse Loader	Python 3.11	Data and Telemetry	45%	the daily load completes before 06:00 local time at every region	the telemetry ingest path, the order broker and the billing usage meter	one transformation still runs on a deprecated warehouse function
Billing Usage Meter	Go 1.22	Data and Telemetry	79%	metered usage reconciles to the order journal within 0.5 per cent each month	the order broker and the WMS connector	none identified
Alarm Router	Python 3.11	Field Reliability	51%	a dispatched alarm reaches an on-call responder inside 90 s at p95	the fleet health monitor and the on-call rotation source	the paging provider's older application interface is still in use
Video Snippet Service	Go 1.22	Applications	43%	a requested snippet is available to the operator inside 45 s	the on-robot camera buffer and the identity service	a transcoder build with known but low-severity upstream advisories
Speed and Separation Monitor	C++17	Safety Engineering	87%	commanded speed is reduced before the separation distance falls below the protective distance	the costmap fusion pipeline and the safety-rated scanner inputs	none; the sensing hardware is current production
Wheel Odometry Filter	C++17	Robot Firmware	62%	odometry drift stays under 1.5 per cent of distance travelled on a clean floor	the wheel encoders and the inertial measurement unit	none identified
Payload Scale Interface	C++17	Robot Firmware	36%	payload mass is reported within 2 per cent of the reference load	the load cell hardware	none identified
Edge Gateway Agent	Rust 1.79	Platform Infrastructure	74%	no telemetry is lost for an outage shorter than four hours	the on-robot services that emit telemetry	none; the transport library was updated in early 2026
Release Orchestrator	Go 1.22	Developer Productivity	55%	no artefact reaches production	the build system, the simulation harness gate and	none identified

Service	Language	Owning team	Coverage	Service-level objective	Upstream dependencies	End-of-life exposure
				without a recorded approval and a passing gate set	the identity service	

5. Safety-relevant incidents and protective-stop behaviour

Grading incident INC-2025-118 on evidence rather than on consequence gives strong, since the reviewer reproduced the failure mode in a laboratory environment. The engineering team's root cause for Identity and Authorization incident INC-2025-118 is that a debug logging statement in the connector build printed the full authorization header, and the reviewer found the analysis sound. INC-2025-118 classifies as a secret exposure event against Identity and Authorization. Scope of impact in secret exposure incident INC-2025-118 was no production site and no robot directly, the exposure being to internal systems, taken from the fleet records rather than from the narrative. Secret exposure event INC-2025-118 is not read as an isolated failure: an operational workaround masked the defect for longer than the incident record suggests. internal incident INC-2025-118 illustrates why TDR-2026-22 separates the severity of a finding from the confidence behind it.

Contributing to INC-2025-118, and not repaired by the remediation, is that build logs were retained far longer than the token lifetime, widening the exposure window. The immediate event in internal incident INC-2025-118 was that a service token with a twelve-hour lifetime was found in a build log retained for ninety days, dated 2025-09-17. Secret exposure incident INC-2025-118 is recorded as follows on delivery: redaction landed on 2025-09-19; retention was changed on 2025-09-24. How incident INC-2025-118 surfaced matters to the assessment: the dependency and secret scanner flagged it on the following scan. The commitment register shows secret exposure event INC-2025-118 against an undertaking to redact authorization headers at the logging layer and cut build log retention to fourteen days, with Sonia Krall named and 2025-10-31 committed. INC-2025-118 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Ironwood Grocery Group incident INC-2025-103 began on 2025-08-22 when two robots repeatedly yielded to each other in a narrow aisle and both stalled for nineteen minutes. As to whether the algorithm defect event INC-2025-103 remediation actually landed: the tie-break landed on 2025-10-19; the scenario was reinstated on 2025-09-02. Incident INC-2025-103 is not read as an isolated failure: an operational workaround masked the defect for longer than the incident record suggests. The two durations recorded for Ironwood Grocery Group incident INC-2025-103 are 19 m to mitigate and 58 days to resolve. Underneath incident INC-2025-103 lies the design decision that the yield heuristic had no tie-break and both robots computed the same deference score. algorithm defect event INC-2025-103 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Ironwood Grocery Group incident INC-2025-103 was detected because the queue depth alert introduced after the March incident fired correctly. Algorithm defect is the class of incident INC-2025-103; Motion Planner is the component. Contributing to INC-2025-103, and not repaired by the remediation, is that the simulation scenario for this geometry existed but had been excluded from the nightly suite. Following incident INC-2025-103 the team undertook to add a deterministic tie-break and return the pinch-point scenario to the nightly suite, with Priya Raghunathan accountable by 2025-10-31. Ironwood Grocery Group felt incident INC-2025-103, which affected one site and 2 robots. Ironwood Grocery Group incident INC-2025-103 was checked against the corresponding entry in the company's own register, and the register was current.

The trigger recorded for Dunbar Industrial incident INC-2026-057 on 2026-05-19 was that state-of-health estimates on opportunity-charged packs drifted enough to hide a cell imbalance until derating. For modelling event INC-2026-057 the reviewer holds strong on the cause and limited on the contributing factor evidence and reports the finding at that strength. Mitigation of modelling incident INC-2026-057 took not a service outage; robots derated and full resolution took open at the review date. Reported detection applies to INC-2026-057, since a robot derated mid-shift and the pack was pulled for analysis. The blast radius of modelling event INC-2026-057 was 2 sites and 5 robots. incident INC-2026-057 is included because a reader of TDR-2026-22 should be able to disagree with the inference without repeating the work.

What modelling incident INC-2026-057 tells the reviewer is that the incident is unremarkable in itself and revealing about the observability of the surrounding system. INC-2026-057 produced one commitment: to refit the estimator on opportunity-charge data and add an imbalance alert independent of state of health, owned by Peter Halvorsen, dated 2026-09-30. Dunbar Industrial incident INC-2026-057 was caused by a design gap: the estimator was fitted on packs that follow a full charge cycle. Contributing to Dunbar Industrial incident INC-2026-057, and not repaired by the remediation, is that the two affected sites moved to opportunity charging after the estimator had been validated. Battery Management Service accounts for INC-2026-057, a modelling event. modelling event INC-2026-057 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

The commitment register shows INC-2025-096 against an undertaking to replace the mailbox fallback with a secondary responder and import rotations from the scheduling system, with Wanda Njoku named and 2025-10-15 committed. INC-2025-096 carries moderate, resting on the team's own post-mortem and one supporting log extract evidence, and the reviewer declined to raise the finding beyond what that supports. The reviewer checked the incident INC-2025-096 commitment against the code and release records; the fallback change landed on 2025-08-31; the rotation import was deferred to 2026. The contributing factor recorded against process gap event INC-2025-096 is that the rotation source of truth was a spreadsheet maintained outside the alarm router. Internal incident INC-2025-096 classifies as a process gap event against Alarm Router. process gap incident INC-2025-096 illustrates why TDR-2026-22 separates the severity of a finding from the confidence behind it.

The detection path for Alarm Router incident INC-2025-096 was discovered the next morning when the alarm was found unacknowledged. The trigger recorded for Alarm Router incident INC-2025-096 on 2025-08-05 was that an overnight alarm routed to a shared mailbox because the on-call rotation had a two-hour gap. Process gap incident INC-2025-096 is not read as an isolated failure: the blast radius was set by a deployment topology decision made years earlier. The affected population in process gap incident INC-2025-096 was no production site and no robot directly, the exposure being to internal systems. Mitigation of internal incident INC-2025-096 took not applicable; no service impact and full resolution took 26 days. process gap incident INC-2025-096 is the entry that changed most between the first and second document requests behind TDR-2026-22.

Grading Localization Engine incident INC-2026-004 on evidence rather than on consequence gives moderate, resting on the team's own post-mortem and one supporting log extract. Scope of impact in performance event INC-2026-004 was one site and 18 robots, taken from the fleet records rather than from the narrative. Underneath performance event INC-2026-004 lies the design decision that the fallback scan matcher is slower than the primary and is selected whenever the map cache misses. Mitigation of incident INC-2026-004 took not a service outage; 5 days to a workaround and full resolution took open at the review date. Performance incident INC-2026-004 records that on 2026-01-13, relocalisation after a lift-and-place exceeded the operator prompt timeout at a newly surveyed site. Kestrel Medical Supply incident INC-2026-004 is the clearest instance in TDR-2026-22 of engineering practice being better than the documentation suggests.

Performance event INC-2026-004 reached the engineering team when the site's commissioning engineer raised it during acceptance testing. Performance is the class of Kestrel Medical Supply incident INC-2026-004; Localization Engine is the component. In incident INC-2026-004, the site was surveyed with a sparse feature set that increases cache misses, which is a process finding rather than a code finding. The reviewer checked the performance incident INC-2026-004 commitment against the code and release records; not landed; the matcher replacement is in design and the re-survey is scheduled. The commitment register shows Localization Engine incident INC-2026-004 against an undertaking to replace the fallback matcher and re-survey the affected aisles, with Rahul Menon named and 2026-06-30 committed. Kestrel Medical Supply incident INC-2026-004 was checked against the corresponding entry in the company's own register, and the register was current.

In incident INC-2025-139 the first signal was the traffic manager's own latency alert fired first, and the reviewer treats that as alerted detection. Capacity event INC-2025-139 was raised on 2025-11-02 after a telemetry backlog on a shared topic delayed control events by up to eleven seconds. Capacity event INC-2025-139 was contained to 7 sites and 246 robots. After capacity event INC-2025-139, Erik Palmgren took the action to split control traffic onto a dedicated topic with its own quota and revisit the topic inventory quarterly by 2026-01-31. The reviewer treats one contributing factor in incident INC-2025-139 as the more important lesson: the topic inventory had recorded the sharing as acceptable

in 2023 and it was never revisited. capacity incident INC-2025-139 is the reason one section of TDR-2026-22 concludes narrowly rather than generally.

Capacity event INC-2025-139 traces back to a single fact: control and telemetry traffic shared one topic and one consumer group budget. Mitigation of Event Bus incident INC-2025-139 took 2 h 47 m and full resolution took 63 days. Event Bus owns Event Bus incident INC-2025-139, which the reviewer classes as capacity. Grading capacity event INC-2025-139 on evidence rather than on consequence gives strong on the cause and limited on the contributing factor. The reviewer's reading of INC-2025-139 is that management's severity rating is defensible and its remediation priority is not. capacity incident INC-2025-139 is the entry that changed most between the first and second document requests behind TDR-2026-22.

1 h 52 m elapsed before INC-2025-041 was mitigated, with resolution at 9 days. Classifying Crestline Fulfillment incident INC-2025-041 as resource leak matters because the Traffic Manager backlog contains two further items of the same kind. Assessed on its own record, Crestline Fulfillment incident INC-2025-041 shows that the remediation addresses the immediate cause and leaves the contributing factor in place. Incident INC-2025-041 touched 2 sites and 61 robots, and Crestline Fulfillment was the affected account. Crestline Fulfillment incident INC-2025-041 was caused by a design gap: the reservation lease had no expiry, so release depended on the departing robot. INC-2025-041 is the clearest instance in TDR-2026-22 of a control that exists on paper and is weaker in practice.

Remediation for Traffic Manager incident INC-2025-041 is to give every corridor reservation a lease with an expiry and add a site-level queue depth alert; Owen Brantley owns it and the committed date is 2025-05-30. INC-2025-041 would have been shorter or smaller had it not been that the alert rule for queue depth existed but was scoped to a metric the site did not emit. Resource leak incident INC-2025-041 rests on limited; the log retention window had closed before the reviewer asked for the raw data evidence, which the reviewer states separately from the severity of the finding. On 2025-03-14 the record for Crestline Fulfillment incident INC-2025-041 opens: a robot lost its network link mid-traverse and its corridor reservation was never released. Reported detection applies to Traffic Manager incident INC-2025-041, since a site supervisor telephoned the on-call engineer after robots queued at one intersection. resource leak event INC-2025-041 was excluded from an earlier draft of TDR-2026-22 and reinstated after the reviewer obtained the supporting record.

As to whether the INC-2025-078 remediation actually landed: landed on 2025-07-28; the regression test was still marked flaky at review. Mitigation of stale state event INC-2025-078 took 1 h 15 m and full resolution took 34 days. INC-2025-078 touched 4 site or tenant with no robot taken out of service, and Harborline Systems was the affected account. Reading stale state incident INC-2025-078 from its first entry, on 2025-06-24 the console listed robots that had been removed from a site and an operator commanded a decommissioned unit. INC-2025-078 belongs to the stale state group and sits with Remote Operations Console. Harborline Systems incident INC-2025-078 is the entry that changed most between the first and second document requests behind TDR-2026-22.

Stale state event INC-2025-078 traces back to a single fact: the session list cached the site roster and only invalidated it on a page reload. The corrective action for stale state event INC-2025-078 is to invalidate the roster cache on a fleet membership event and add a session-level regression test, and 2025-08-31 is the date the team gave. Stale state incident INC-2025-078 carries strong, resting on the production traces the team supplied evidence, and the reviewer declined to raise the finding beyond what that supports. In incident INC-2025-078 the first signal was an internal engineer noticed the discrepancy during a demonstration, and the reviewer treats that as reported detection. Contributing to Harborline Systems incident INC-2025-078, and not repaired by the remediation, is that no test covered a roster change during an open operator session. stale state event INC-2025-078 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2026-22)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-320 pallet mover	depth camera	3.9.3	11	both rigs	pre-production
HR-320 pallet mover	navigation lidar	4.4.6	8	neither rig	current
HR-200 tote mover	network radio	1.0.0	3	both rigs	one revision behind
HR-200 tote mover	load cell deck	1.11.6	5	neither rig	superseded
HR-320 pallet mover	compute module	4.12.4	2	rig one only	one revision behind
HR-320 pallet mover	E-stop harness	4.17.0	13	neither rig	one revision behind
HR-300 pallet mover	wheel encoder set	3.1.8	6	rig two only	current

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-320 pallet mover	wheel encoder set	2.8.4	13	neither rig	current
HR-410 manipulator	network radio	3.3.3	9	both rigs	pre-production
HR-410 manipulator	safety scanner	1.3.4	2	rig two only	one revision behind
HR-200 tote mover	navigation lidar	1.4.0	6	neither rig	superseded
HR-300 pallet mover	navigation lidar	1.0.6	8	rig two only	superseded
HR-320 pallet mover	depth camera	2.13.7	10	both rigs	current
HR-410 manipulator	E-stop harness	3.16.5	2	rig one only	one revision behind
HR-320 pallet mover	wheel encoder set	3.9.3	2	rig one only	current
HR-200 tote mover	inertial measurement unit	2.6.1	14	rig two only	one revision behind
HR-320 pallet mover	safety scanner	3.10.7	9	neither rig	current
HR-300 pallet mover	navigation lidar	2.13.1	9	rig two only	pre-production
HR-300 pallet mover	navigation lidar	1.5.2	4	both rigs	current
HR-300 pallet mover	compute module	1.7.4	12	rig one only	pre-production
HR-240 tote mover	compute module	4.17.3	10	both rigs	pre-production
HR-200 tote mover	load cell deck	2.15.7	9	rig two only	one revision behind

5. Safety-relevant incidents and protective-stop behaviour - part 2 of 2

Below, the 5. Safety-relevant incidents and protective-stop behaviour continues; nothing separates this part from the last except the page on which it happens to fall. Pagination alone determines where it falls.

Column definitions stated with the first part of the 5. Safety-relevant incidents and protective-stop behaviour govern the whole of it, and any qualification recorded there is carried forward here. The sequence is unbroken either side of it.

Schedule O - Robot variant, module revision and bench rig coverage (TDR-2026-22) (continued, part 2 of 2)

Robot variant	Module	Firmware revision	Sites deployed	Bench rig coverage	Revision status
HR-410 manipulator	load cell deck	3.11.3	2	rig two only	current
HR-320 pallet mover	gripper coprocessor	4.12.6	5	rig two only	pre-production
HR-410 manipulator	navigation lidar	1.2.2	10	rig one only	current
HR-300 pallet mover	E-stop harness	1.16.4	14	rig one only	superseded
HR-320 pallet mover	charge contact assembly	4.2.7	13	rig one only	current
HR-240 tote mover	network radio	1.6.8	12	rig one only	one revision behind
HR-200 tote mover	compute module	4.16.3	11	rig one only	one revision behind
HR-410 manipulator	network radio	4.6.3	3	rig one only	current
HR-410 manipulator	load cell deck	3.13.2	8	neither rig	superseded
HR-320 pallet mover	compute module	3.17.0	10	rig two only	superseded
HR-320 pallet mover	charge contact assembly	3.12.5	6	both rigs	superseded
HR-240 tote mover	wheel encoder set	1.2.1	4	rig two only	one revision behind
HR-300 pallet mover	E-stop harness	1.12.8	10	neither rig	pre-production
HR-200 tote mover	compute module	3.19.5	9	rig one only	superseded
HR-410 manipulator	inertial measurement unit	3.9.1	5	rig one only	pre-production
HR-320 pallet mover	wheel encoder set	4.13.0	5	neither rig	pre-production
HR-300 pallet mover	gripper coprocessor	1.5.6	11	rig two only	pre-production
HR-200 tote mover	E-stop harness	3.0.2	14	neither rig	current
HR-320 pallet mover	inertial measurement unit	4.18.4	3	rig two only	one revision behind
HR-410 manipulator	gripper coprocessor	4.9.7	13	neither rig	pre-production
HR-320 pallet mover	load cell deck	3.5.8	2	rig one only	superseded
HR-320 pallet mover	load cell deck	2.3.6	9	rig one only	one revision behind

6. Verification method and evidence behind each control

Residual risk at hazard HZ-20 in the register is stated as Tolerable where cleaning compliance is evidenced, and the status is partially verified. Lidar contamination degrades detection without a fault being raised is the hazard entered at Serious injury hazard HZ-20. HZ-20 is relevant to the sites that have moved to opportunity charging rather than to the whole fleet. hazard HZ-20 in the register was cross-checked against the engineering record supplied for TDR-2026-22, and the two agree.

HZ-04 was scoped to robots fitted with the weighing deck, and the reviewer checked the scoping against the fleet record. After mitigation, Remote-exposure hazard HZ-04 is Tolerable once calibration is gated in the service workflow;

the register status is verified. Mitigation for hazard HZ-04 is force limiting with a mechanical release and an interlock on the access panel. Serious injury hazard HZ-04 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Status verified and residual risk Tolerable with the measured approach speed assumption stated are what the register gives for HZ-01. Hazard HZ-01 in the register identifies the hazard that robot contacts a pedestrian while travelling in a shared aisle. For Serious injury hazard HZ-01 the verifying activity was witnessed approach testing on six robots plus a review of the on-robot safety log. Frequent-exposure hazard HZ-01 was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Fire hazard HZ-10 was verified through supplier test report plus an internal witnessed disconnect test. Entry Fire hazard HZ-10 in the hazard register reads: charging contact arcs during an unplanned disconnect. The population exposed to HZ-10 is the newest three sites, where commissioning is incomplete. Improbable-exposure hazard HZ-10 is the clearest instance in TDR-2026-22 of engineering practice being better than the documentation suggests.

The record shows that Fire hazard HZ-11 produced one field event that did not result in harm. The parameter set for Fire hazard HZ-11 reads Fire, Improbable, Low. Improbable-exposure hazard HZ-11 remains Broadly acceptable, subject to the state-of-health drift item after the control is applied, and the register marks it partially verified. register entry HZ-11 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Hazard HZ-03 is not clean: robots without the weighing deck rely on operator loading discipline. The residual entry for Occasional-exposure hazard HZ-03 reads Tolerable where the weighing deck is fitted, partially verified. Serious injury hazard HZ-03 is mitigated by payload mass and centre-of-gravity limits enforced by the scale interface. HZ-03 would be a candidate for the first post-close engineering review under any reasonable plan.

Schedule M - Remediation commitment tracker with evidence of delivery (TDR-2026-22)

Commitment	Source	Owner	Committed date	Status	Evidence inspected
give every corridor reservation a lease with an expiry and add a site-level queue depth alert	INC-2025-041	Owen Brantley	2025-05-30	landed	runbook change inspected
re-derive the time offset on every reconnect and add the affected variant to the second bench ri	INC-2025-047	Yuki Harada	2025-06-15	partly landed	runbook change inspected
release charge windows on a shift-end event and reconcile the calendar nightly	INC-2025-052	Nadia Semenova	2025-06-30	landed	test added and observed
key suppression on the tenant and identifier pair and backfill the affected order records	INC-2025-058	Tomas Villareal	2025-07-15	landed	release record inspected
make the allocator consult charger availability before excluding a robot	INC-2025-063	Marta Kellerman	2025-07-31	landed	test added and observed
add a leadership and under-replicated partition check to the broker upgrade runbook	INC-2025-071	Erik Palmgren	2025-07-31	landed	commit inspected
invalidate the roster cache on a fleet membership event and add a session-level regression test	INC-2025-078	Anjali Sekhon	2025-08-31	landed	configuration export inspected
persist the pause state and require a design note for any change to the release state machine	INC-2025-084	Erik Palmgren	2025-08-15	landed	no evidence located

Commitment	Source	Owner	Committed date	Status	Evidence inspected
add the connector torque check to incoming inspection and retrofit the affected harness batch	INC-2025-090	Bettina Vogel	2025-09-30	partly landed	configuration export inspected
replace the mailbox fallback with a secondary responder and import rotations from the scheduling	INC-2025-096	Wanda Njoku	2025-10-15	partly landed	release record inspected
add a deterministic tie-break and return the pinch-point scenario to the nightly suite	INC-2025-103	Priya Raghunathan	2025-10-31	partly landed	configuration export inspected
fail the load on an incomplete partition and reprocess the affected days	INC-2025-110	Farida Cisse	2025-10-31	landed	interview only
redact authorization headers at the logging layer and cut build log retention to fourteen days	INC-2025-118	Sonia Krall	2025-10-31	partly landed	test added and observed
annotate staleness on every read path and add a rebalance-under-load test to the platform suite	INC-2025-124	Marta Kellerman	2025-11-30	landed	commit inspected
restrict export to published revisions and shorten the publication path so the workaround is unn	INC-2025-131	Lucia Ferrante	2025-12-15	partly landed	release record inspected
split control traffic onto a dedicated topic with its own quota and revisit the topic inventory	INC-2025-139	Erik Palmgren	2026-01-31	partly landed	no evidence located
fold the reset into the re-run command and make double-counting detectable in the ledger	INC-2025-146	Nadia Semenova	2025-12-31	landed	configuration export inspected
apply the lifecycle rule through the configuration baseline and delete the over-retained clips	INC-2025-152	Anjali Sekhon	2026-01-31	partly landed	runbook change inspected
replace the fallback matcher and re-survey the affected aisles	INC-2026-004	Rahul Menon	2026-06-30	not landed	release record inspected
require a calibration record before returning a manipulator robot to service	INC-2026-009	Ines Duarte	2026-04-15	landed	test added and observed
move precondition validation to apply time and add an offline apply case to the platform suite	INC-2026-015	Sam Oyelaran	2026-03-31	landed	configuration export inspected
pin the full build configuration and add a determinism check to the harness	INC-2026-021	Colin Achebe	2026-05-31	not landed	test added and observed
group on the signal	INC-2026-028	Wanda Njoku	2026-04-30	landed	commit inspected

Commitment	Source	Owner	Committed date	Status	Evidence inspected
and block closure while a constituent alert remains active					
measure closing rate where the sensing permits and re-run the protective distance calculation	INC-2026-033	Bettina Vogel	2026-08-31	not landed	runbook change inspected
serialise cancellation against assignment and add an hourly orphan sweep	INC-2026-039	Grace Lindqvist	2026-05-31	landed	configuration export inspected
aggregate buffer pressure to a site alert and extend the buffer to twelve hours on the current h	INC-2026-045	Sam Oyelaran	2026-06-30	partly landed	configuration export inspected
check every template against the baseline and re-converge the four forked templates	INC-2026-051	Marcus Duplessis	2026-06-15	partly landed	configuration export inspected
refit the estimator on opportunity-charge data and add an imbalance alert independent of state o	INC-2026-057	Peter Halvorsen	2026-09-30	not landed	commit inspected
require a justification and a second approver within one working day for every emergency release	INC-2026-062	Colin Achebe	2026-08-31	not landed	test added and observed
add a slip detector that flags the condition and issue a site advisory on cleaning windows	INC-2026-068	Yuki Harada	2026-09-15	not landed	release record inspected
Partition the fleet allocator write path	TD-01	engineering leadership	2027-03-31	in design	register entry only
Replace the unmaintained scan-matching library in the localization fallback	TD-02	engineering leadership	2026-12-31	committed	register entry only
Retire the vendored geometry library in costmap fusion	TD-03	engineering leadership	2027-06-30	backlog	register entry only
Raise robot software line coverage above sixty per cent on the perception path	TD-04	engineering leadership	2027-03-31	in progress	register entry only
Bring the second bench rig to the current sensor configuration	TD-05	engineering leadership	2026-09-30	committed	register entry only
Make the nightly simulation suite blocking	TD-06	engineering leadership	2026-11-30	blocked	register entry only

7. Safety-related engineering work not yet funded

The plan treats programme item TD-20 as funded in the current financial year, which the reviewer checked against the headcount model. Reading TD-20 against the funded plan, the prototype uses a network-copyleft component that must be replaced before shipping. The consequence of leaving TD-20 on Wheel Odometry Filter undone is that it drift on wet floors is neither corrected nor flagged. Wheel Odometry Filter item TD-20 appears twice in the working papers for TDR-2026-22 because the first answer was corrected after a follow-up request.

Reading programme item TD-06 against the funded plan, blocked on the physics core determinism work. As at the review date register item TD-06 was blocked, with 2026-11-30 still the stated target. Register item TD-06 is sequenced

behind a supplier release outside the company's control. TD-06 on Simulation Harness appears twice in the working papers for TDR-2026-22 because the first answer was corrected after a follow-up request.

The reviewer's qualification on safety evidence item TD-18 is that the safety team supports it; it is not funded in the current plan. Effort on programme item TD-18 is estimated at 0.75 engineer-quarters, costed at $104,000. Safety Controller item TD-18 carries a 2027-03-31 target and a status of backlog. register item TD-18 has no single accountable owner, and that is the finding the reviewer would press hardest.

Under security supportability, entry TD-08 is to replace the archived charting library in the operator console. About register item TD-08 it should be said that no advisory affects the current version, which is why it ranks low. Funded from the second quarter onward is the funding position on TD-08. security supportability item TD-08 has no single accountable owner, and that is the finding the reviewer would press hardest.

On funding, configuration item TD-11 is not funded, and carried in the register regardless. Programme item TD-11 is in progress with a target date of 2026-09-30. Two templates remain forked as at the review date is the point the reviewer would put to management on configuration item TD-11. TD-11 was tested rather than accepted, and the test is described in the working papers behind TDR-2026-22.

Quality item TD-22 cannot start earlier than it does because it is ahead of two other items that depend on it. The plan treats Payload Scale Interface item TD-22 as funded only if the team's headcount plan is met, which the reviewer checked against the headcount model. For TD-22 the team estimates 0.25 engineer-quarters, which the reviewer converted to $42,000 for comparison. quality item TD-22 is recorded here without adjustment; the reviewer saw no basis to restate what the company reported.

Behind a customer maintenance window that is not yet agreed is where programme item TD-24 sits in the programme. IP hygiene item TD-24 carries a 2026-09-30 target and a status of committed. Funded in the current financial year is the funding position on TD-24 on Costmap Fusion. programme item TD-24 has no single accountable owner, and that is the finding the reviewer would press hardest.

Requires capital for a third rig rather than engineering effort is the point the reviewer would put to management on delivery item TD-14. 2027-03-31 is the target for programme item TD-14, whose status is backlog. The estimate for TD-14 is 0.5 engineer-quarters and $88,000. register item TD-14 appears twice in the working papers for TDR-2026-22 because the first answer was corrected after a follow-up request.

Left in place, the condition behind TD-01 caps the region at roughly eight sites at current order density. Sequencing of TD-01 matters: it is last in a three-item chain, so its date is the least reliable on the register. Scalability item TD-01 carries a 2027-03-31 target and a status of in design. TD-01 illustrates why TDR-2026-22 separates the severity of a finding from the confidence behind it.

Unfunded and dependent on capital rather than engineering effort is the funding position on TD-17. The register entry security item TD-17 reads: enforce a two-person rule on OTA artefact signing, in the security category. The consequence of leaving security item TD-17 undone is that it a single key holder can sign firmware. security item TD-17 is reported at the evidence strength available and has not been promoted to fit the narrative.

The reviewer's comment on programme item TD-03 is that management ranked this behind the separation monitoring work. The estimate for programme item TD-03 is 1 engineer-quarter and $128,000. The register entry TD-03 on Costmap Fusion reads: retire the vendored geometry library in costmap fusion, in the support category. support item TD-03 is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

What register item TD-07 protects against is the fact that it non-reproducible test results undermine every simulation claim. The register entry Simulation Harness item TD-07 reads: pin the full simulation build configuration and enable a determinism check, in the tooling category. Funded in the current financial year is the funding position on tooling item TD-07. Simulation Harness item TD-07 was excluded from an earlier draft of TDR-2026-22 and reinstated after the reviewer obtained the supporting record.

Status on TD-12 on Edge Gateway Agent is in progress; the target is 2026-10-31. On funding, TD-12 on Edge Gateway Agent is unfunded and dependent on capital rather than engineering effort. Register item TD-12 covers the work

to extend the edge buffer to twelve hours. TD-12 was one of two items where TDR-2026-22 could not obtain the underlying data within the engagement.

8. Safety-rated components and third-party assessment

The open question on pointcloud-utils is that file-level copyleft; three modified files must be published and the reviewer could not evidence publication. Tracing pointcloud-utils to its source gives public registry, signature verified. In Costmap Fusion the component pointcloud-utils is used point cloud filtering in fusion, which is the fact the licence analysis rests on. pointcloud-utils was compared with the equivalent position at two other components, and it is neither the best nor the worst.

Policy compliance for scanmatch-legacy rests on its approval entry, which is not approved. Scanmatch-legacy was assessed for support as well as licence risk, and it carries a single-maintainer dependency with no succession plan. Linking mode matters for scanmatch-legacy, which is used linked into the localization fallback path. scanmatch-legacy is the only entry in its group where the reviewer and the engineering team reached different conclusions.

Licensing for grpc-transport is Apache-2.0 and the copyleft column reads no. Grpc-transport presents a component with open advisories that do not affect the deployed configuration to the engineering team. OSS-004 records grpc-transport 1.62.0 against Event Bus, service-to-service transport across the cloud. grpc-transport was verified to the extent the available evidence permits, and the residual uncertainty is stated rather than absorbed.

The supply route for chartkit is public registry; upstream archived in 2025. Chartkit appears at OSS-006 in version 4.8.2; the usage recorded is operator console visualisations. No copyleft applies to chartkit, whose licence is BSD-3-Clause. chartkit is one of the entries where TDR-2026-22 relies on the company's own measurement rather than an independent one.

In the approval log, sqlite-embedded is approved 2023-06-15. The open question on sqlite-embedded is that no attribution obligation. Edge Gateway Agent is where sqlite-embedded is consumed, which determines whether distribution occurs at all. sqlite-embedded is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

Usage of jwt-verify 5.2.1 is token verification in the identity service, and the inventory reference is OSS-014. In addition to its terms, jwt-verify brings a build-time dependency whose absence would stop the pipeline rather than the product. Provenance of jwt-verify is public registry, signature verified. jwt-verify was reviewed against the position twelve months earlier, and nothing measurable has changed.

Attribution present is the reviewer's observation on csv-export. Public registry, signature verified is the provenance recorded for csv-export. Csv-export carries a practical obligation: to reproduce the notice text in the delivered documentation. csv-export is the clearest instance in TDR-2026-22 of a control that exists on paper and is weaker in practice.

Where image-signing came from matters here: public registry, signature verified. Image-signing also carries no identified risk beyond ordinary version currency, which is a supportability question rather than a licence one. Approval for image-signing reads approved 2025-01-15, and the reviewer checked the approving record. image-signing is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Opencv-subset sits inside Gripper Controller and is therefore in the shipped product rather than in tooling. On provenance, opencv-subset shows public registry, signature verified. Meeting the terms on opencv-subset means to attribute the component in the customer-facing notice file. opencv-subset is the entry that changed most between the first and second document requests behind TDR-2026-22.

Licensing for physics-core-fork is Zlib and the copyleft column reads no. Beyond licensing, physics-core-fork carries an upstream project that is active but changes interface frequently. Meeting the terms on physics-core-fork means to share alike any derived asset it distributes. physics-core-fork was excluded from an earlier draft of TDR-2026-22 and reinstated after the reviewer obtained the supporting record.

The inventory entry OSS-022 is http-router 1.9.4, cloud service routing. The terms applying to http-router are MIT; the copyleft classification is no. What the company must actually do about http-router is to exclude a particular build

configuration from the shipped artefact. http-router illustrates why TDR-2026-22 separates the severity of a finding from the confidence behind it.

9. Security findings with a safety consequence

Medium-severity finding SEC-2025-07 is not fully retired by its remediation: medium; two forked templates were not re-issued until May 2026. The reason assessment finding SEC-2025-07 was not downgraded is that the same secret was reusable at any site provisioned from that template. Substantiating SEC-2025-07 is the fact that the secret was recovered from a template export, graded limited, resting on a configuration read rather than on a demonstrated exploitation. On finding SEC-2025-07 the conclusion is that the finding was closed within the committed window and the closure evidence was inspected. assessment finding SEC-2025-07 was reviewed against the position twelve months earlier, and the direction of travel is favourable.

SEC-2025-06 produces this judgement: the retest evidence is a screenshot rather than a re-run of the original test, which the reviewer notes. SEC-2025-06 against Video Snippet Service records that video snippet index allowed cross-tenant enumeration by identifier, discovered on 2025-09-19. As at the review date assessment finding SEC-2025-06 was remediated and the closure entry reads 2025-12-18. The practical consequence in finding SEC-2025-06 is that a customer operator could enumerate the existence of clips in another tenant. SEC-2025-06 against Video Snippet Service is not material on its own and belongs in this report because of what it reveals about the surrounding process.

Read alongside the remediation record, SEC-2026-05 against Alarm Router shows that the remediation is adequate and the time to remediate is not. The reason assessment finding SEC-2026-05 was not downgraded is that one leaked credential affects paging for every region. Assessment finding SEC-2026-05 leaves residual risk that the reviewer assesses as medium; per-region credentials are scheduled but not issued. Finding SEC-2026-05 rests on the observation that credential reuse confirmed from the secret store inventory; the reviewer holds that to be moderate. Alarm Router finding SEC-2026-05 was checked against the company's own register, which had not been updated since the change landed.

The evidence for Low-severity finding SEC-2025-11 is moderate: attribute absence confirmed; no working exploit chain was demonstrated. On 2025-09-25 the tester recorded assessment finding SEC-2025-11: console session cookie lacked a strict same-site attribute. Remote Operations Console finding SEC-2025-11 is not fully retired by its remediation: low. Remote Operations Console finding SEC-2025-11 stands remediated; 2025-10-30 is what the record gives for closure. Low-severity finding SEC-2025-11 is the entry that changed most between the first and second document requests behind TDR-2026-22.

Assessment finding SEC-2026-04 matters because customer commercial data is retained in operational logs. SEC-2026-04 is assessed as follows: the exposure is internal rather than customer-facing, which lowers the severity without removing the finding. Reading SEC-2026-04, the defect is that wMS connector logged customer order payloads at debug level; the discovery date is 2026-04-16. Residual risk after remediation of finding SEC-2026-04 is low; payload logging requires an explicit per-tenant approval. assessment finding SEC-2026-04 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

The remediation ledger shows Low-severity finding SEC-2026-12 as partially remediated, pending. SEC-2026-12 was opened on 2026-04-20 and states that edge gateway agent retained mutual authentication keys after decommissioning. Weighing SEC-2026-12 against Edge Gateway Agent against the compensating controls, the item is open past its committed date and no revised date has been agreed. In SEC-2026-12 against Edge Gateway Agent the exploitation path means a disposed compute module carries a usable fleet credential. Edge Gateway Agent finding SEC-2026-12 sits inside the sample rather than being an exception to it, which is the reason it is reported at length.

Low-severity finding SEC-2026-20 is rated Low with a numeric band of 3.2 on the tester's scoring scale. The reviewer declines to soften SEC-2026-20: the underlying pattern appears in two other components that the tester did not reach. Not planned is the closure position on finding SEC-2026-20, which the ledger records as accepted. A calibration change cannot be reconstructed during an investigation is the exposure SEC-2026-20 against Payload Scale Interface describes. SEC-2026-20 against Payload Scale Interface is one of the items the reviewer would put on a monitoring pack rather than on a closing checklist.

The residual position on Low-severity finding SEC-2025-10 is low, stated after mitigation rather than assuming full success. Low is the rating SEC-2025-10 against Charge Scheduler carries, at 3.7 on the assessor's own scale. 2025-10-09 is the closure position on assessment finding SEC-2025-10, which the ledger records as remediated. Strong evidence supports assessment finding SEC-2025-10, namely that endpoint reachable and content captured. finding SEC-2025-10 is included because a reader of TDR-2026-22 should be able to disagree with the inference without repeating the work.

Assessment finding SEC-2026-02 sits against Robot Shadow Store and records that robot shadow store cache lacked authorization on the internal read path. At 7.6 the score on finding SEC-2026-02 places it in the High group. The reviewer's position on assessment finding SEC-2026-02 is that the compensating control is real but depends on a network boundary that a single misconfiguration removes. Residual risk after remediation of SEC-2026-02 against Robot Shadow Store is low; authorization is checked on both the cache and durable paths. SEC-2026-02 against Robot Shadow Store appears twice in the working papers for TDR-2026-22 because the first answer was corrected after a follow-up request.

Assessment finding SEC-2026-06 rests on the observation that role policy read from the cloud configuration export; the reviewer holds that to be strong. The reason Medium-severity finding SEC-2026-06 was not downgraded is that a compromised scenario could read production artefact storage. SEC-2026-06 against Simulation Harness reduces to a residual risk of medium; the role was scoped down but still reads one production bucket. On 2026-04-17 the tester recorded Simulation Harness finding SEC-2026-06: simulation harness cluster used a long-lived cloud role with broad read access. assessment finding SEC-2026-06 was the subject of a specific management representation, which the reviewer has recorded but not relied on alone.

Remediation status on SEC-2026-09 is remediated; the record shows 2026-05-08. Finding SEC-2026-09 reduces to a residual risk of low. Read alongside the remediation record, SEC-2026-09 against Fleet Health Monitor shows that the reviewer would not treat this as a closing condition and would put it on the first post-close sprint. Rating for finding SEC-2026-09 is Low, band 3.9; the reviewer re-read the scoring rationale and accepted it. finding SEC-2026-09 has an owner who accepted the finding immediately, which is worth recording alongside the finding itself.

10. Open safety qualifications carried forward

Brandywine Technology Partners will not treat an open item in the present report as closed on a verbal assurance, and has said so to the engineering leadership. Open items in report TDR-2026-22 are stated with what would close each one, so a post-close plan can be costed against them. One declinature in the present report rests on an engineering judgement the reviewer shares; the others rest on prioritisation, which is a different kind of argument. Where management has declined to act, report TDR-2026-22 preserves the decision and the stated rationale so a post-close plan can revisit it. Brandywine Technology Partners has flagged 2 matters for the transaction record and left the remainder as engineering work.

Management's declinatures are reported in report TDR-2026-22 at face value; the reviewer's disagreement is stated separately and does not overwrite them. TDR-2026-22 treats a backlog label as evidence of intent and not as evidence of remediation, which is why several items remain open in this report that the company regards as handled. 9 items in TDR-2026-22 are open, two of which the reviewer would treat as pre-closing information requests rather than post-close work. Three of the open items in the present report would close on a single document that the company has agreed to produce. Any figure in the present report that the reviewer could not trace to a system is attributed to the person who gave it.

Where this report says a defect is verified, the reviewer reproduced it or read a production trace of it; everything else is recorded as a concern. Grading in this report is deliberately coarse, because a finer scale would imply a precision the underlying evidence does not carry. In the present report the reviewer distinguishes what was verified from what was represented, and marks each conclusion accordingly. Report TDR-2026-22 follows the reviewer's usual practice of grading evidence strength separately from finding severity, so a plausible concern is not promoted to a verified defect. Readers comparing this report with an earlier technical report should note that the service inventory was renamed in between.

TDR-2026-22 recommends that the remediation commitment register be reported to the board quarterly with evidence attached, not just status. Brandywine Technology Partners recommends that the 5 indicators be reported with their measurement basis attached, since two of them changed basis during the reviewed period. N. Thackeray did not visit

the two most recently commissioned sites, so the commissioning quality there is taken from the company's records alone. Where the report at TDR-2026-22 reports a supplier's test result, the reviewer read the certificate and did not witness the test. Cross-references in report TDR-2026-22 point to the evidence index rather than to a summary of the evidence.

11. Conclusions on the safety position

The report at TDR-2026-22 concludes that the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. N. Thackeray would summarise TDR-2026-22 in one sentence: the principal risk is concentrated in a small number of components on the control path, which makes the mitigation tractable. Brandywine Technology Partners notes that 2 of the matters in TDR-2026-22 are capable of being closed before completion if the company is asked in time. For the transaction, this report identifies 2 matters the reviewer would raise before closing and 4 for the post-close engineering plan. TDR-2026-22 was prepared on the footing that a reader may be an engineer or may not, and defines terms where the distinction matters.

The costed remediation implied by TDR-2026-22 is $955,000 over the coming four quarters, which is within the ordinary engineering budget. 4 items in the report at TDR-2026-22 are post-close engineering work rather than transaction issues, and the reviewer has costed them at $955,000. Brandywine Technology Partners asked the same four questions of every team interviewed for report TDR-2026-22, which is how the inconsistencies below were found. Interviews for report TDR-2026-22 were conducted with the owning engineer present rather than with a manager speaking on their behalf. This report is one input to a diligence process and is not a substitute for the company's own engineering judgement.

This report proposes 5 metrics for the post-close monitoring pack, each already produced by an existing system. Reporting under report TDR-2026-22 should distinguish a target that was met from a target that was lowered, which the current pack does not. 5 declined items are carried in the report at TDR-2026-22, of which two lack any documented compensating control. Not every finding in TDR-2026-22 is accepted by management: 5 are declined, and the declinatures are reproduced rather than argued away. Brandywine Technology Partners has costed remediation at the company's own loaded engineering rate rather than at a market rate.

For Telemetry Ingest the stated target is that telemetry visible in the operator console inside 30 s of emission at p95, expressed against production traffic rather than a benchmark. For Telemetry Ingest the reviewer records that management's own assessment is broadly supported, with the coverage gap treated as a genuine finding rather than a presentational one. End-of-life review of Telemetry Ingest found the stream processor runs a release that leaves community support in late 2026. Farida Cisse leads Data and Telemetry's work on Telemetry Ingest and was the reviewer's principal interview for it. Telemetry Ingest sits between the edge gateway agent and the event bus and the fleet health monitor, the analytics warehouse loader and the alarm router, and that position is what makes its availability objective load-bearing. Telemetry Ingest is reported here in full because a summary of it in an earlier draft of TDR-2026-22 was misread as an assurance.

Taking the evidence together, WMS Connector the component is adequate today and carries a supportability liability that is not on the funded plan. Service-level intent for WMS Connector states that order acknowledgement returned to the warehouse system inside 1.5 s at p95. Data ownership is clear for WMS Connector: it writes PostgreSQL 14 plus a message outbox table and no other service writes there. The ageing position for WMS Connector is one customer integration still runs on a connector build from 2024, and the reviewer checked it against the vendor statements on file. The topology for WMS Connector is cloud, one deployment per customer tenant, and the reviewer confirmed it against the deployment manifests. WMS Connector was discussed with two engineers who gave slightly different accounts, and the difference is preserved here.